PHASESET: PERMUTATION-INVARIANT PERIODIC RELATION TOKENS FOR MULTI-PERSON MOTION–LANGUAGE RETRIEVAL

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This manuscript registers a method and evaluation protocol only. No training run, test score, confidence interval, or submission authority is represented.

ABSTRACT

Group motions can share actor marginals and the same multiset of pair relations while differing in which relations meet at each actor. We register PhaseSet, a permutation-invariant residual for bidirectional multi-person motion–language retrieval. Six fixed Morlet bands produce directed pair descriptors; symmetric pair statistics and actor–relation incidence moments yield six group tokens. The topology branch is exactly zero at two actors, and all edges are processed with deterministic streamed reductions. The confirmatory protocol fixes a 572-capture Embody 3D eligibility census, a participant-component-disjoint 400/96/76 split, unseen-cardinality K=3 testing, 33 training attempts, eight hypotheses, one sealed test, and 100,000 clustered bootstrap draws. This is a **NO_RESULT** technical draft: every empirical cell is held.

Index Terms— multi-person motion, text–motion retrieval, periodic relations, set representation

1. PROBLEM AND CLAIM BOUNDARY

Single-person retrieval already includes contrastive motion synthesis and fine-grained wavelet encoders [1, 2]. Dyadic motion work covers interaction-aware retrieval and frequency-guided editing [3, 4]. Multi-person language systems cover open-domain generation and instruction-based understanding [5, 6, 7]. Periodic video-language reasoning also appears in action counting [8]. Consequently, PhaseSet makes no broad "first" claim for motion–language, frequency-aware retrieval, or multi-human understanding.

The sole registered question is narrower: *does frequency-specific actor–relation incidence structure add retrieval utility beyond a qualified group encoder and seven matched structural controls?* The contribution is a fixed specification plus a fail-closed experiment. It is not generation, editing, causal phase inference, role recognition, or an official reimplementation of another model. Language supplies semantic retrieval tokens but cannot generate, select, edit, or supervise physical phase, lag, power, or coherence fields.

2. RELATED EVIDENCE

Embody 3D provides synchronized multi-person motion, audio, and text at a larger official scale than the eligibility subset used here [9]. Multi-TPC offers natural triadic conversation, whereas M3Act3D offers synthetic large-K group activity; we reserve them for cross-domain and engineering probes, respectively [10, 11]. Neither is confirmatory evidence for the present retrieval claim. SocialGen/SocialX and Multi-Motion support variable-person modeling, but their autoregressive/generative tasks and constructed data do not replace native full-gallery retrieval [6, 5].

The five supplied reference papers further delimit the task. MotionMaster is single-person generation/editing; LangPose uses language as a pose-lifting pretraining prior; HMVLM is a single-person multi-task language model; and Pose2Lang3D distills 3D cues into static 2D-pose VQA [12, 13, 14, 15]. They motivate lineage, masking, task-interference, and privileged-information audits, not a PhaseSet result. Cross-wavelet analysis itself is longstanding [16]; our fixed descriptors are pooled motion features, not novel signal-processing estimators.

3. PHASESET

3.1. Input and deterministic periodic relations

An input is a dynamically padded tensor of $K \geq 2$ actors, 200 frames, and 22 body joints, with actor/joint/frame masks. A shared 512D actor encoder uses no actor ID. Motion is resampled to 20 Hz; an independently hashed 20-Hz Morlet bank retains PhasePair's six physical center frequencies while its legacy 30-Hz tap bytes remain unchanged. Each response is computed once per actor. For every valid unordered pair $\{i, j\}$ and band b, the inherited directed 13D descriptor $d_b(i \to j)$ contains log auto-powers for the ordered endpoints, coherence-like magnitude, cosine/sine of principal relative phase, wrapped delay, a signed-phase-valid bit, and a six-way band indicator. These are motion-derived descriptors, not causal or physical measurements.

All pairs are visited lexicographically in canonical 64-edge microblocks. The implementation allocates no tensor with two actor axes. Pair processing is $O(K^2)$ but streamed; if all exact edges exceed the frozen resource envelope, the API returns `RESOURCE_LIMIT` instead of dropping actors or sampling edges.

3.2. Pair and incidence paths

A shared half-edge network and a symmetric pair network compute

$$
\begin{aligned}
u^b_{ij} &= H(d_b(i \to j)), \quad u^b_{ji} = H(d_b(j \to i)), \\
p^b_{ij} &= P\Big(u^b_{ij} + u^b_{ji}, |u^b_{ij} - u^b_{ji}|, u^b_{ij} \odot u^b_{ji}\Big).
\end{aligned}
\tag{1}
$$

Ordered concatenation is prohibited. The pair component $\bar{p}_b$ is the canonical masked mean over p^b_{ij}. For actor i, incident half-edges produce mean offset from the band-global mean, population second central moment $M2$, coverage $\deg(i)/(K-1)$, and missingness. The topology NodeMLP input is exactly $[\Delta\text{mean}, M2, \text{coverage}, \text{missing}]$. A node is valid only at degree at least two; the post-MLP hard gate has no biased layer after it. Thus

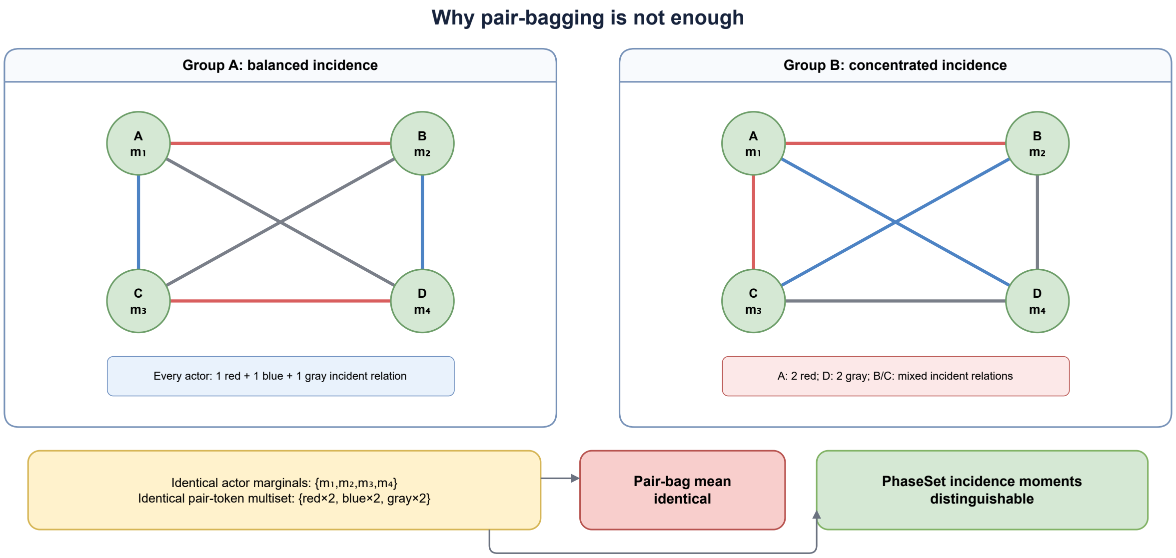

Fig. 1. Conceptual counterexample (not data or a result). Actor marginals and the pair-token multiset are identical, so pair-bag averaging cannot distinguish the groups; incident relation moments retain their different ownership.

at $K=2$ the topology residual and its parameter gradients are exact positive zero.

Let q_b be the canonical masked mean of valid node outputs. The band token is

$$g_b = m_b\, C(\bar{p}_b + \tanh(\lambda_b) q_b)\,, \tag{2}$$

where m_b is the band-valid mask and C_b is the common postprocessor. The capacity-matched pair-only control passes exact positive zero for q_b through the same C_b. Fixed-tree float64 microblock reduction, compensated microblock accumulation, and fixed-order Chan merges make the output invariant to registered chunk sizes.

3.3. Text alignment and objective

CLIP's frozen text encoder supplies an embedding [17]. A shared TextBandMLP and learned band ID yield text tokens ℓ_b. For capture x and text y,

$$S(x, y) = S_{\text{base}}(x, y) + \tanh(\alpha) \frac{\sum_b m_b \cos(g_b, \ell_b)}{\sum_b m_b}. \tag{3}$$

The externally supplied base scores are frozen. Variable-positive symmetric InfoNCE treats every registered caption for one capture as positive; windows or captions never become independent statistical units.

4. FROZEN DATA PROTOCOL

The primary source is native multi-person Embody 3D, subject to its access and non-redistribution terms [9]. The project-registered, public-safe eligibility census has 572 captures, 69 participants, 20.673 scene hours, and 16 participant co-occurrence components: 27 captures have $K=3$ and 545 have $K=4$. The split is 400 train / 96 validation / 76 test captures and 48/10/11 participants, with neither participant nor capture overlap. Crucially, *all* 27 $K=3$ captures are in test; $K=3$ is therefore an unseen-participant and unseen-cardinality slice. We must not imply that training saw $K=3$.

Motion uses 22 SMPL-X body joints, antialiased 30-to-20 fps resampling, and nonoverlapping 10 s (200-frame) windows. The common origin is the first valid frame's all-group pelvis centroid; centering/orienting around an indexed actor is forbidden. Whole-group yaw augmentation preserves relations. Gaps up to 0.25 s may be interpolated while retaining the missing mask; a window is rejected if any actor is below 95% valid or has a longer gap. Coherence never deletes an edge.

Primary test queries are human holistic capture descriptions. Every valid nonoverlapping window is encoded and pooled by a fixed masked mean. Captures without holistic text are removed before scoring, never reassigned. A separate 10 s auxiliary task may use visibly machine-fused actor text; its model is barred from skeletons, spectra, phase, outputs, and split statistics. The same-family semantic-review status is `PROVISIONAL_SAME_MODEL_FAMILY_REVIEW_NOT_GATE_CLOSING` and closes no data, execution, sealed-test, or scientific-claim gate.

5. REGISTERED EXPERIMENTS: ALL RESULTS HELD

The base candidates are ActorMean (B0), SetPMA (B1), and SocialTemporal (B2). Each uses 512D states, eight heads, a 2048D FFN, dropout 0.1, and seeds 1729, 2718, and 31415: nine qualification attempts. The winner is chosen only by three-seed mean validation bidirectional R@1, with ties resolved by fewer parameters, lower frozen-runtime latency, then smaller ID. Its seed-specific checkpoints are frozen and shared across residual systems. Every one of the nine rows binds the same validation manifest, query census, and evaluator code, plus unique terminal, checkpoint, and score-artifact digests; a trusted host must authenticate that exact row census before qualification is accepted.

B0 applies one shared temporal actor tower and a masked actor mean. B1 replaces that mean with permutation-invariant pooling by

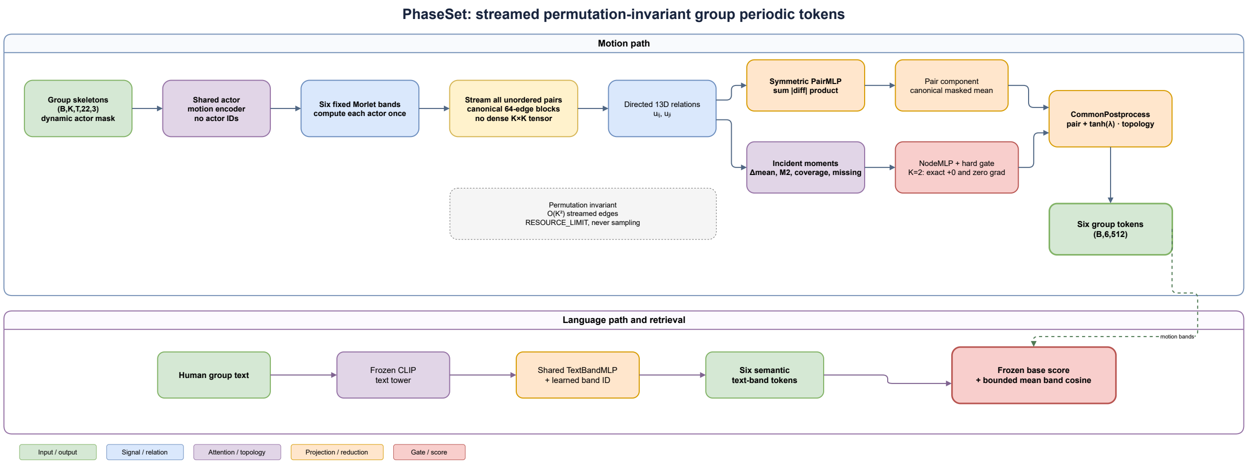

Fig. 2. Registered architecture (conceptual, no-result). Directed relations feed a symmetric pair mean and incident-moment topology path; a hard $K=2$ gate preserves dyadic equivalence. Six motion tokens align with six semantic text-band tokens on top of the frozen qualified base score.

multihead attention (PMA). B2 uses four alternating blocks: temporal self-attention is applied independently with shared weights, then each frame performs set attention across the valid actors, followed by PMA. No candidate receives an actor index, actor-specific tower, or actor-0 reference. The architecture decision uses the three-seed validation mean, not per-seed winners; an exact tie is resolved only by the registered resource order. Thus every residual system for a seed reads the same immutable winning-base checkpoint and cache.

Systems 01–08 each run at the same three seeds (24 more attempts), for a formal total of **33 training attempts**. Bases use the registered AdamW schedule (learning rate 2×10^{-4}, 30 epochs); residuals use 3×10^{-4} and 20 epochs. Comparable systems share precision, batch, data, captions, and checkpoint rules and remain within $\pm 1\%$ of full trainable parameters.

The effective global batch is 128 windows under an edge-count budget, using gradient accumulation when cardinality reduces the microbatch. Both schedules use AdamW, weight decay 0.01, 5% linear warmup then cosine decay, and unit gradient clipping. FP32 is the default. BF16 is admitted only after the target GPU passes the frozen numerical suite, and one precision is shared by all comparators. Checkpoint selection reads only the fixed validation endpoint; test access cannot change an epoch, prompt, cache, or system definition.

5.1. Capacity-matched structural controls

The controls separate token capacity, marginal periodicity, spectral content, pair aggregation, missingness, incidence ownership, and signed phase. Generic six-token residuals (01) have no motion-derived periodic field. Marginal power (02) removes cross-person relations; mean/difference DCT (03) supplies a non-phase spectral control. Pair-bag (04) retains every symmetric pair token but discards edge ownership. Coverage-only (05) receives coverage and missingness without half-edge content. Incidence-shuffled (06) preserves the half-edge multiset, degree, and coverage while reassigning edges to actors. Phase-stripped full (07) preserves energy and coherence but removes signed phase and lag. Only full PhaseSet (08) retains both pair content and true incidence. Controls share the same postprocessor and, where possible, the same head with inputs disabled rather than a separate lower-capacity network.

5.2. Execution and scaling gates

The runtime computes one Morlet response per actor and band, then checkpoints edge chunks so backward recomputes rather than stores all pair activations. The live footprint is $O(KD + E_c D)$ for chunk size E_c, while exact work is $O(K^2)$. Batches are bucketed by $\sum_b \binom{K_b}{2}$. Canonical private commitments determine reduction order but never enter the neural network. Permutation, padding, mixed-batch, endpoint-swap, chunk, and differentiable descriptor-seam gradient-equivariance tests are mandatory before real data. Stress cases at $K \in \{8, 32, 128, 256\}$ must show no allocation with two actor axes; an insufficient edge budget yields an explicit resource failure rather than a different graph.

The primary endpoint is the equal-weight average of T2M group R@1 and M2T any-positive R@1. Secondary endpoints include bidirectional R@1/3/5/10, median rank, K=3, K=4, macro-K, parameters, FLOPs, peak memory, throughput, and K scaling. H1 (08 vs 00) is the sole primary hypothesis. H2–H8 compare 08 with 01–07, respectively: generic tokens, marginal power, DCT, pair-bag, coverage-only, incidence-shuffled, and phase-stripped controls. H2–H8 form one Holm family; H1 is reported separately.

Group-Hard-≤ 32 is auxiliary and never changes a group cardinality. It first enforces the frozen duration, per-actor marginal-band-power, total-energy, and root-speed calipers, then ranks eligible negatives by frozen text similarity. Because the sealed K=3 gallery contains only 27 captures, its strictly K-matched gallery uses all eligible K=3 candidates up to the 32-candidate cap; it never inserts a K=4 negative or duplicates a capture. Insufficient caliper-matched negatives are an evaluation failure, not a reason to widen the rule after seeing scores. A deterministic count-only diagnostic must be constant within every such gallery; its reference chance is therefore the reciprocal of that query's actual gallery size. Scorer independence is established only by an externally authenticated pre-score freeze binding over its code/checkpoint, exact similarity digest, excluded 27-checkpoint census, manifests, and final gallery. A numerical ranking heuristic alone is not treated as independence evidence.

After all validation choices, caption/gallery digests, code, and aggregation are frozen, the protocol admits exactly one sealed-test execution. Every one of the 9×3 system/seed tables is ranked inde-

Table 1. Frozen full-gallery matrix; every result is HOLD.

ID	System	1729	2718	31415	Aggregate
00	Qualified group base	HOLD	HOLD	HOLD	HOLD
01	Generic six tokens	HOLD	HOLD	HOLD	HOLD
02	Marginal Morlet power	HOLD	HOLD	HOLD	HOLD
03	Mean/difference DCT	HOLD	HOLD	HOLD	HOLD
04	PhasePair pair-bag	HOLD	HOLD	HOLD	HOLD
05	Coverage/missing-only	HOLD	HOLD	HOLD	HOLD
06	Incidence-shuffled	HOLD	HOLD	HOLD	HOLD
07	Phase-stripped full	HOLD	HOLD	HOLD	HOLD
08	PhaseSet full	HOLD	HOLD	HOLD	HOLD

Table 2. Frozen slice/scaling report; dash is out of scope.

K	Source / role	Full	Hard-$\leq$ 32	Memory	Throughput
3	Embody / unseen-cardinality	HOLD	HOLD	HOLD	HOLD
4	Embody / seen-cardinality	HOLD	HOLD	HOLD	HOLD
8	M3Act3D + synthetic / scale	–	–	HOLD	HOLD
16	M3Act3D / group-size	–	–	HOLD	HOLD
27	M3Act3D / large-group	–	–	HOLD	HOLD
32	synthetic / edge stress	–	–	HOLD	HOLD
128	synthetic / edge stress	–	–	HOLD	HOLD
256	synthetic / resource limit	–	–	HOLD	HOLD

pendently; logits are never averaged across seeds. For each capture and comparison, the three paired seed effects are averaged in fixed order (1729, 2718, 31415), after which 100,000 shared paired bootstrap draws resample captures only. Seeds are fixed repeated blocks, not resampled or independent units; captions and windows are not units either. The three test participant components also receive leave-one-component-out sensitivity estimates. Test data cannot select a model, checkpoint, seed, hyperparameter, prompt, gallery, metric, or claim. Failed/interrupted attempts remain in the 33-run census, and failed hypotheses remain reportable.

For every H1–H8 comparison, the seed-blocked paired contribution is formed at capture level after keeping all captions and windows together. Exactly the same 100,000 bootstrap index draws produce the point difference, percentile interval, and two-sided tail probability. H1 is reported without multiplicity adjustment; the seven raw probabilities for H2–H8 enter one deterministic Holm step-down family. The three participant components in test are also removed one at a time and the full aggregate is recomputed. This sensitivity analysis is descriptive and cannot replace the preregistered full-gallery decision. The aggregate binds all 27 checkpoint, execution, score-table, and rebuilt evaluation-report digests. Per-system cells summarize the arithmetic mean and population standard deviation of three independent seed metrics, never a score from an averaged-logit matrix.

Claim language is generated by a fixed decision table rather than written from the most favorable cell. Full must beat both pair-bag and incidence-shuffled before an incidence-topology statement is allowed; it must beat phase-stripped before a signed-phase statement is allowed; and beating the base without beating generic tokens is reported only as a capacity-compatible gain. Any null, reversal, failed run, or resource limit remains in the terminal ledger. One content-addressed aggregate renders every numeric JSON/CSV and LaTeX table, preventing a manuscript number from diverging from the released machine-readable result. The editable conceptual figures and slides contain no empirical values.

6. CONCLUSION AND LIMITS

At present there is no empirical conclusion. If H1 later passes, the maximum base-level statement is an incremental change on the exact frozen gallery and protocol. “Topology-supported” additionally requires stable superiority to both pair-bag (H5) and incidence-shuffled (H7); “phase-supported” requires superiority to phase-stripped (H8); “structure-supported” requires superiority to generic tokens (H2). Otherwise the report must descend to base-improvement-only, null, or negative. No outcome establishes causal coordination, universal group understanding, intent/identity/ability inference, generation/editing ability, encoder equivariance, or out-of-domain generalization.

The fixed bands, pooled phase descriptors, only 27 zero-shot K=3 captures, three test participant components, and conditional frozen-gallery bootstrap limit inference. Human-motion records and text require license, privacy, and publication review; this repository exposes no raw motion, captions, participant identities, reversible memberships, or private model artifacts.

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